

Kabir Roy

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EDUCATION

VIT Bhopal University

B.Tech in Computer Science and Engineering — CGPA: 9.40

Bhopal, India

Aug 2023 – May 2027

EXPERIENCE

Defence Research and Development Organisation (DRDO)

Software Engineering Intern

New Delhi, India

May 2025 – Jun 2025

- Deployed a multi-host Hyperledger Fabric blockchain network across 2 Google Cloud VMs (Org1/Org2), using Docker Swarm to orchestrate distributed containers for peers, orderers, and CAs.
- Automated cryptographic identity generation (cryptogen) and secure cross-VM certificate/chaincode distribution via SSH key-based auth and SCP to enable multi-organization channel setup.
- Managed the full chaincode lifecycle — package, install, approve, commit, and query — with CouchDB state-database integration, validating asset creation and query transactions across peer organizations.

Madhya Pradesh Police — Cybercrime & Data Analysis Division

Data Analyst Intern

Bhopal, India

Sept 2025 – Dec 2025

- Analyzed 1,000+ cybercrime records using Python and SQL to identify fraud patterns across multiple case categories, improving detection accuracy for the division.
- Cleaned and structured raw case data in Pandas and compiled summary breakdowns in Excel to help investigators track recurring fraud trends and flag high-risk cases faster.

PROJECTS

AsthmaAI: Hybrid Asthma Risk Prediction System | Python, XGBoost, LightGBM, SHAP | GitHub

2025

- Engineered a 6-model stacking ensemble with entropy-gated clinical safety heuristics to classify asthma risk into High/Medium/Low tiers.
- Validated the system on two independent clinical datasets, achieving 92.57% accuracy (1,010 patients) and 96.45% accuracy (2,392 patients).
- Applied SHAP-based explainability and demographic fairness auditing across age and gender subgroups to ensure interpretable, equitable predictions.

Manufacturing Operations & OEE Analytics Dashboard | Python, Pandas, Power BI, DAX | GitHub

2026

- Architected a Python/Pandas ETL pipeline transforming raw hourly machine logs into a Kimball-style star schema (1 fact table, 5 dimension tables) with enforced data-integrity constraints.
- Designed a Power BI dashboard with custom DAX measures for real-time OEE, Month-over-Month tracking, and Mean Time Between Failures (MTBF) across production lines.
- Built a downtime Pareto analysis and shift-performance comparison (Morning/Afternoon/Night) to surface the small set of failure causes driving most production loss.

CERTIFICATIONS

- AWS Certified Cloud Practitioner** — Amazon Web Services (AWS), 2026
- Introduction to Machine Learning** — NPTEL, Jan – Apr 2025
- Google Analytics Certification** — Google, Issued May 2025

TECHNICAL SKILLS

Languages: C++, Python, SQL

Data Science: XGBoost, LightGBM, SHAP, Scikit-learn, Power BI, DAX, NumPy, Pandas, Matplotlib, Seaborn

Cloud & Tools: AWS (Cloud Practitioner), Git, GitHub, Android Studio, Jupyter Notebook

Databases: MySQL, SQL

Concepts: RESTful APIs, Data Analytics, ETL, Star Schema Design

LEADERSHIP & EXTRACURRICULAR

Bit by Bit Club — Vice President

Former Student Coordinator

VIT Bhopal University

2026

- Hosted large-scale coding events with 200+ participants, boosting club outreach by 50%; promoted to VP for leadership excellence.
- Organized 3+ technical workshops on AI, Machine Learning, and Web Development.
- National CyberShield Hackathon** — Top 10 of 750+ teams, 2025.
- MP Police Cybersecurity Hackathon** — Finalist, 2025.